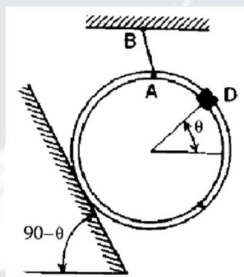
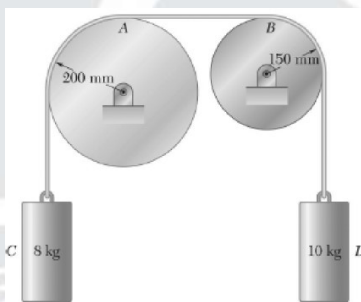


1. A ring is shown supported by wire AB and a smooth surface. The ring has a mass  $m$  and a mean radius  $r$ . A body D having a mass of  $\frac{m}{2}$  is fixed to the ring as shown. If the wire is severed, what is the acceleration of body D at this instant ?

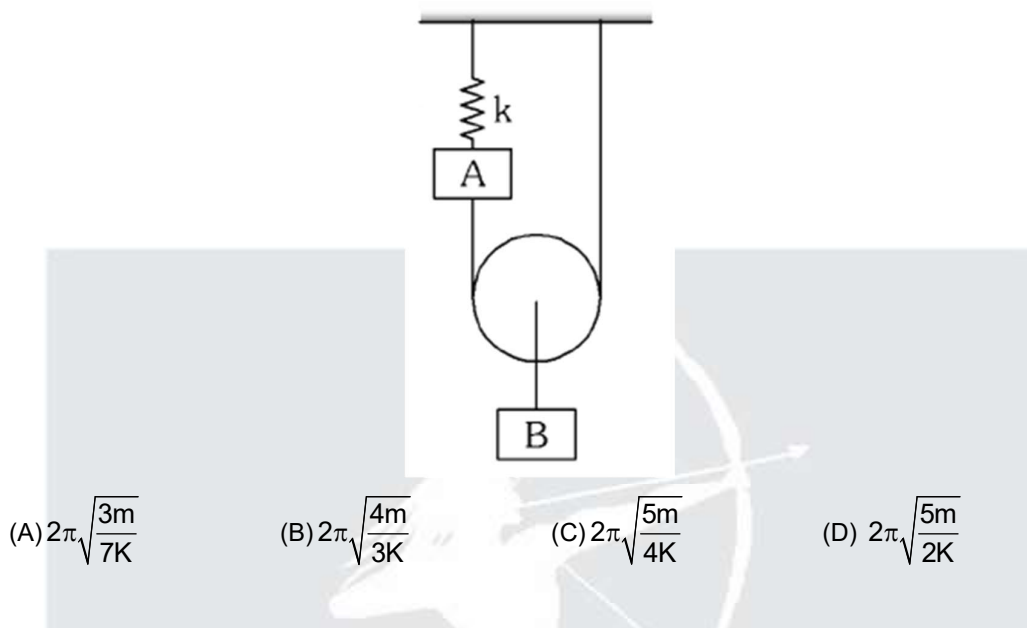


- (A)  $g \cos \theta$       (B)  $g \sin \theta$       (C)  $\frac{g \sin \theta}{3}$       (D)  $\frac{g \sin \theta}{4}$
2. Two uniform disks and two cylinders are assembled as indicated. Disk A has a mass of 10 kg and disk B has a mass of 6 kg. Knowing that the system is released from rest, the time required for cylinder C to have a speed of 0.5 m/s. (The cylinders are attached to a single cord that passes over the disks. Assume that no slipping occurs between the cord and the disks.)



- (A) 2.3 sec      (B) 0.66 sec      (C) 0.15 sec      (D) 1.7 sec

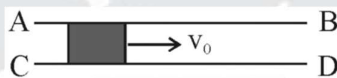
3. In the system shown in figure, Both the blocks A and B have equal mass of  $m$  and the force constant of the ideal spring is  $k$ . Pulley and threads are massless. The time period of vertical oscillations of block A is:



4. Four identical samples of same polyatomic ideal gas undergo addition of same amount of heat under four different processes namely  $PV^{1/2} = C$ ;  $PV^{3/2} = C$ ;  $P = C$  and  $PV^2 = C$ . The final internal energy will be maximum for the sample which has undergone

- (A)  $PV^{1/2} = C$  (B)  $PV^{3/2} = C$  (C)  $P = C$  (D)  $PV^2 = C$

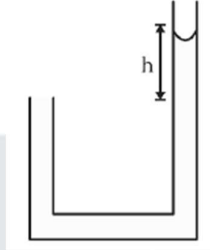
5. In the figure, AB and CD are two vertical walls of a horizontal rectangular groove. A rectangular block of mass  $m$  just fits in the groove. Base of the block is smooth and coefficient of friction between walls and block is  $\mu$ . Contact area of block with each wall is  $A$ . Linear coefficient of thermal expansion and young's modulus of material of block are  $\alpha$  and  $Y$  respectively. Temperature  $\theta$  of block increases uniformly as  $A\theta = t$  ( $t$  = time). Initial velocity of block is  $v_0$  then



- (A) Retardation of block at time  $t$  is  $\frac{\mu YA \alpha t}{m}$  (B) Velocity at time  $t$  is  $\left( v_0 - \frac{\mu YA \alpha t^2}{m} \right)$

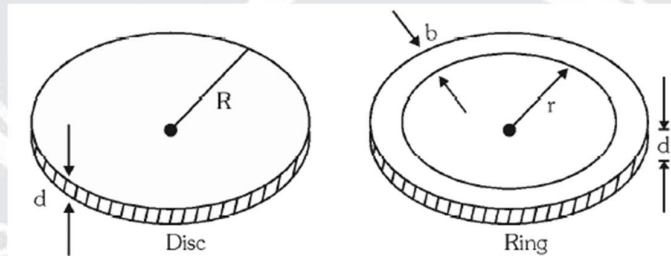
- (C) Time taken by block to stop is  $\sqrt{\frac{mv_0}{\mu YA \alpha}}$  (D) work done by friction is  $\frac{1}{2}mv_0^2$

6. Two capillaries of small cross section are connected as shown in the figure. The right tube has cross sectional radius  $R$  and left one has a radius of  $r (< R)$ . The tube of radius  $R$  is very long whereas the tube of radius  $r$  is of short length. Water is slowly poured in the right tube. Contact angle for the tube wall and water is  $\theta = 0^\circ$ . Let  $h$  be the height difference between water surface in the right and left tube. Surface tension of water is  $T$  and its density is  $\rho$ .



Choose the correct statement(s):

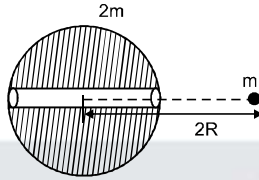
- (A) The maximum value of  $h$  for which water will not flow out of left tube is  $\left(\frac{2T}{\rho g}\right)\left(\frac{r+R}{rR}\right)$
- (B) The maximum value of  $h$  for which water will not flow out of left tube is  $\left(\frac{T}{\rho g}\right)\left(\frac{r+R}{rR}\right)$
- (C) The value of  $h$  for which the water surface in left tube is found to be flat is  $\left(\frac{T}{R\rho g}\right)$
- (D) The value of  $h$  for which the water surface in left tube is found to be flat is  $\left(\frac{2T}{R\rho g}\right)$
7. A steel ring is to be fitted on a wooden disc of radius  $R$  and thickness  $d$ . The inner radius of the ring is  $r$  which is slightly smaller than  $R$ . The outer radius of the ring is  $r + b$  and its thickness is  $d$  (same as the disc). There is no change in value of  $b$  and  $d$  after the ring is fitted over the disc ; only the inner radius becomes  $R$ . If the Young's modulus of steel is  $Y$ .



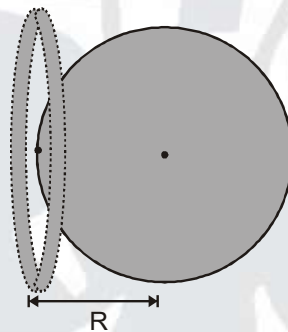
Consider the correct statement(s):

- (A) The strain in the ring is  $\left(\frac{R-r}{r}\right)$
- (B) The tension force developed in the ring is  $\frac{Ybd(R-r)}{r}$
- (C) The tension force developed in the ring is  $\frac{Ybd(R-r)}{3r}$
- (D) The tension force developed in the ring is  $\frac{Ybd(R-r)}{4r}$

8. A solid spherical planet of mass  $2m$  and radius ' $R$ ' has a very thin tunnel along its diameter. A small cosmic particle of mass  $m$  is at a distance  $2R$  from the centre of the planet as shown. Both are initially at rest, and due to gravitational attraction, both start moving toward each other. After some time, the cosmic particle passes through the centre of the planet. (Assume the planet and the cosmic particle are isolated from other planets)

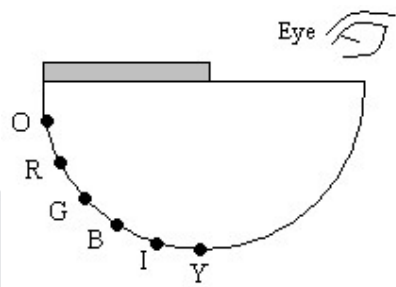


- (A) Displacement of the cosmic particle till that instant is  $\frac{4R}{3}$
- (B) Acceleration of the cosmic particle at that instant is zero
- (C) velocity of the cosmic particle at that instant is  $\sqrt{\frac{8Gm}{3R}}$
- (D) Total work done by the gravitational force on both the particle is  $-\frac{2Gm^2}{R}$
9. An uncharged conducting sphere of radius  $R$  is placed near a uniformly charged ring of radius  $R$ . Total charge on ring is  $Q$ . The centre of sphere lies on the axis of ring and distance of centre of sphere from centre of ring is  $R$ .



- (A) Potential at centre of ring is  $\frac{KQ}{R}$
- (B) Potential at centre of ring is  $\frac{KQ}{\sqrt{2}R}$
- (C) Potential at centre of ring due to conducting sphere is zero
- (D) Potential at centre of conducting sphere is  $\frac{KQ}{\sqrt{2}R}$

10. A person is looking at the flat surface of a transparent hemisphere. Half of the flat surface is coloured black and half of curved surface is coloured in six equal strips as shown in the figure. Then select correct alternative/s ( $n_R = 1.412$ ,  $n_O = 1.413$ ,  $n_Y = 1.414$ ,  $n_G = 1.415$ ,  $n_B = 1.416$ ,  $n_V = 1.417$ ) (where alphabates has usual meaning)



- (A) The person can see green orange and red  
 (B) The person can see only yellow, indigo and blue  
 (C) The ray coming from green, red and orange strips will be totally reflected  
 (D) The ray coming from yellow, indigo and blue strips will be totally reflected
11. Density of a sphere of radius  $R$  with concentric spherical cavity of radius  $r = \frac{2R}{3}$  is given by,

$\rho = \frac{\rho_0}{R}(R - x)$ , where  $x$  is distance from its centre and  $\left(\frac{2R}{3} \leq x \leq R\right)$ , then

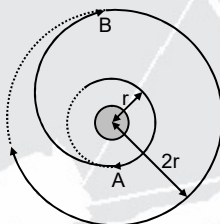
**List-I**

**List-II**

- |     |                                                                                        |     |    |
|-----|----------------------------------------------------------------------------------------|-----|----|
| (P) | If potential on its outer surface $\frac{-a\rho_0 GR^2}{81}$ then $a =$                | (1) | 4  |
| (Q) | Potential at its inner surface $\frac{-b\rho_0 GR^2}{81}$ then $b =$                   | (2) | 11 |
| (R) | Field on its surface $\frac{c\rho_0 GR}{81}$ , then $c =$                              | (3) | 0  |
| (S) | Field at a distance $\frac{R}{2}$ from its centre $\frac{d\rho_0 GR}{81}$ , then $d =$ | (4) | 14 |
- (A)  $P \rightarrow 2$ ;  $Q \rightarrow 4$ ;  $S \rightarrow 2$ ;  $R \rightarrow 3$       (B)  $P \rightarrow 4$ ;  $Q \rightarrow 3$ ;  $S \rightarrow 2$ ;  $R \rightarrow 1$   
 (C)  $P \rightarrow 1$ ;  $Q \rightarrow 2$ ;  $S \rightarrow 3$ ;  $R \rightarrow 4$       (D)  $P \rightarrow 4$ ;  $Q \rightarrow 1$ ;  $S \rightarrow 3$ ;  $R \rightarrow 2$

**Paragraph for Question Nos. 12 to 13**

The Hohmann Transfer Orbit maneuver is used by spaceships to transfer from one circular orbit around a planet to another, by temporarily entering an intermediate elliptical orbit. The spaceship starts on a small circular orbit of radius  $r$  around a planet of mass  $M$ . At point A in the diagram below, it fires its propellers to put itself onto an intermediate elliptical orbit. At point B, it fires its propellers again, and ends on the larger circular orbit of radius  $2r$ . You may assume that the propellers are so powerful that the changes in velocity occur instantaneously at each point. Between point A and B space ship experience only gravitational force from planet.



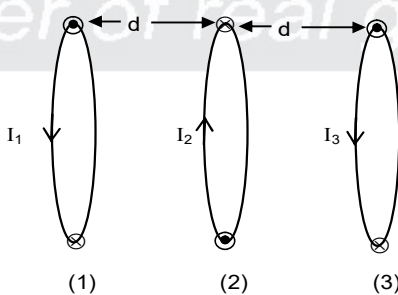
12. Find  $\Delta V_A$  (the amount by which the velocity needs to be changed at point A) needed to achieve this travel path.

(A)  $\sqrt{\frac{GM}{r}} \left( \sqrt{\frac{2}{3}} - 1 \right)$  (B)  $\sqrt{\frac{GM}{r}} \left( \sqrt{\frac{4}{3}} - 1 \right)$  (C)  $\sqrt{\frac{GM}{r}} \left( \sqrt{\frac{4}{3}} + 1 \right)$  (D)  $\sqrt{\frac{2GM}{r}} \left( \sqrt{\frac{4}{3}} - 1 \right)$

13. Time period of a satellite in inner orbit is 2 days then what is find taken by space ship to go from A to B.

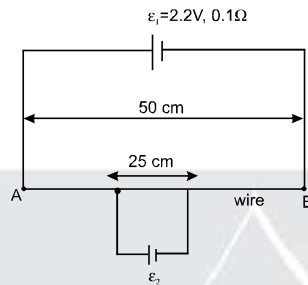
(A)  $(2)^{3/2}$  days (B)  $\left(\frac{4}{3}\right)^{3/2}$  days (C)  $\left(\frac{3}{2}\right)^{3/2}$  days (D)  $\left(\frac{2}{3}\right)^{3/2}$  days

14. Consider three identical conducting loops kept parallel to each other such that their axis coincides. Loop (1), loop (2) and loop (3) self inductance of each loop is  $L$ . Current in loop (1) is  $I_1$ , in loop (2) is  $I_2$  and in loop (3) is  $I_3$ , rate of change in current in them at certain instant is  $\frac{dI_1}{dt}$ ,  $\frac{dI_2}{dt}$ ,  $\frac{dI_3}{dt}$  respectively. Their mutual inductance is  $M_{ij}$  (i.e. for 1 & 2 its  $M_{12}$ ) current in loop (1) and (3) is in same sense and in (2) it is in opposite sense :

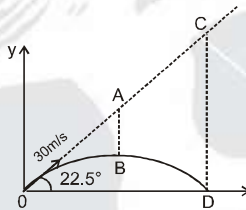


Given that if  $\frac{dI_1}{dt} = \frac{dI_2}{dt} = \frac{dI_3}{dt}$ , emf induced in loop (2) is zero. Find  $\frac{L}{M_{12}}$

15. A thin uniform wire AB of length 50 cm and resistance  $1\Omega$  is connected to the terminals of a battery of emf  $\varepsilon_1 = 2.2\text{ V}$  and internal resistance  $0.1\Omega$ . If the terminals of another cell (assume ideal) are connected to two points 25 cm apart on the wire AB without altering the current in the wire AB, then emf  $\varepsilon_2$  of cell in volts is :



16. A particle is projected with a speed of 30m/s at angle  $22.5^\circ$  with horizontal from ground as shown. If AB and CD are parallel to y-axis and B is highest point of trajectory of particle, then the value of CD/AB is :

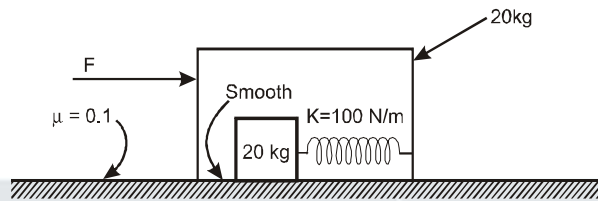


17. A man does 30 J of work on a ball to throw it vertically upto a height  $h$  on earth, if man does work  $x\text{ J}$  on same ball to throw it vertically upto height  $2h$  on another planet having its radius  $\frac{1}{4}$  of radius of earth and planet has same density as that of earth. Find 'x' ?
18. A binary star has a time period 3 years (time period of earth is one year) while distance between these two stars is 9 times distance between the earth and the sun. Mass of one star is equal to mass of the sun and mass of other star is  $5n$  times mass of the sun then calculate  $n$ .
19. A closed steel cylinder is completely filled with water at  $0^\circ\text{C}$ . The water is made to freeze at  $0^\circ\text{C}$ . It is known that density of water at  $0^\circ\text{C}$  is  $1000\text{kg/m}^3$  and the density of ice at  $0^\circ\text{C}$  is  $910\text{kg/m}^3$ . Bulk modulus of ice at  $0^\circ\text{C}$  is nearly  $9 \times 10^9\text{Pa}$ . [Now you can easily understand why pipelines burst in cold regions as the winter sets in.]. The rise in pressure on cylindrical wall  $K \times 10^8\text{pascals}$  is approximately. The value of  $K$  \_\_\_\_\_.
20. In the last question 19 let the length of the tube be  $L$  and its outer radius be  $R$ . Water rises in it to a height  $h$ . Mass of empty tube is  $M$ . The magnitude of vertical force needed to hold the tube in this position is found to be

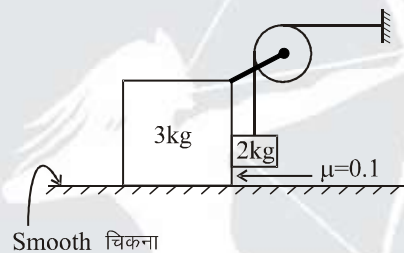
$$F = xMg + yP_0\pi\left[R^2 - \frac{Lr^2}{L-h}\right] + z\pi(R+r)T$$

The value of  $(x + y + z)$  is \_\_\_\_\_.

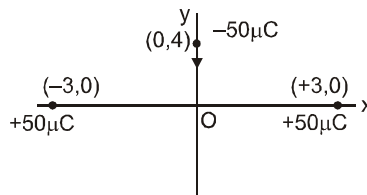
21. A box of mass 20 kg is kept on a rough horizontal surface ( $\mu = 0.1$ ). There is a spring mass system having block of mass 20 kg and spring of spring constant 100 N/m in the box. There is no friction between block and box and spring is initially relaxed. Now an external force of 120 N is applied on the box, find acceleration of the box (in  $\text{m/s}^2$ ) just after applying the force.



22. System shown in figure is released from rest. If the acceleration of 2kg block is  $\frac{25\sqrt{2}}{x} \text{ ms}^{-2}$  then x will be :



23. A particle A is at origin, and another particle B is at  $x = 5\text{m}$  at  $t = 0$  and they are moving with constant velocities  $(2\hat{i} + 2\hat{j} + \hat{k})\text{m/s}$  and  $(-2\hat{i} - \hat{j} + \hat{k})\text{m/s}$  respectively. What is the minimum distance (in m) between these particle during the course of their motion ?
24. Two uniformly charged identical non-conducting rings of radius R are fixed coaxially at separation of  $2R$ . A uniformly charged non-conducting sphere of radius R is placed between rings such that its centre lies at the mid point of line joining the centres of two rings. Charge on ring-1 is Q. Electric field and potential is zero at centre of the sphere. If the potential at the centre of ring-2 is  $\frac{KQ}{R} \left( 1 + \frac{1}{\sqrt{x}} + \frac{2\sqrt{z}}{y} \right)$  then,  $x - y - z$  will be :
25. The figure shows two equal, positive charges, each of magnitude  $50 \mu\text{C}$ , fixed at points  $(3, 0) \text{ m}$  and  $(-3, 0) \text{ m}$  respectively. A charge  $-50 \mu\text{C}$ , moving along negative y-axis has a kinetic energy of 4J at the instant it crosses point  $(0, 4) \text{ m}$ . If the y co-ordinate of this charge where direction of its motion reverses for the first time after crossing this point is  $x\sqrt{2} \text{ m}$  then, x is (neglect gravity).



## ANSWER KEY

|     |      |     |      |     |       |     |      |     |      |
|-----|------|-----|------|-----|-------|-----|------|-----|------|
| 1.  | (A)  | 2.  | (B)  | 3.  | (C)   | 4.  | (B)  | 5.  | (BC) |
| 6.  | (AD) | 7.  | (AB) | 8.  | (ABC) | 9.  | (BD) | 10. | (BC) |
| 11. | (A)  | 12. | (B)  | 13. | (C)   | 14. | 2    | 15. | 1    |
| 16. | 4    | 17. | 15   | 18. | 16    | 19. | 8    | 20. | 4    |
| 21. | 4    | 22. | 9    | 23. | 3     | 24. | +6   | 25. | -6   |

